ETH zürich · Language, Reasoning & Education Lab · ETH AI CENTER

Understanding Failures in LLM Reasoning by Learning Structured Representations of Chain-of-Thought

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TL;DR Reasoning quality lives in the **CoT structure**. A GNN over **CoT dependency graphs**, with **no step text**, predicts correctness, and its **explanations** reveal the **structural patterns** behind failed reasoning.

up to **+7.8%** correctness prediction vs. length baseline

up to **+2.5%** correctness prediction vs. tree-based LCoT2Tree

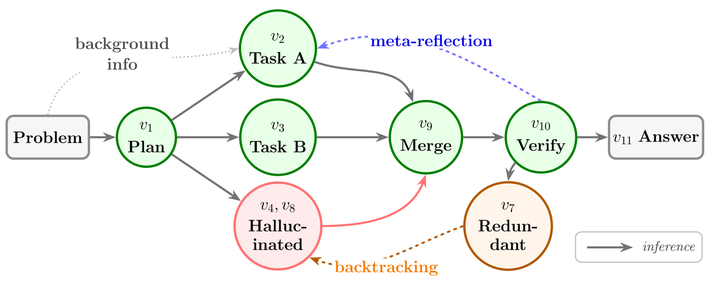
49% of steps are path inefficiency vs. 6.6% verification

56% Best-of-*N*, CoT2Graph + entropy (mean: 43.2%)

1 Why look at structure?

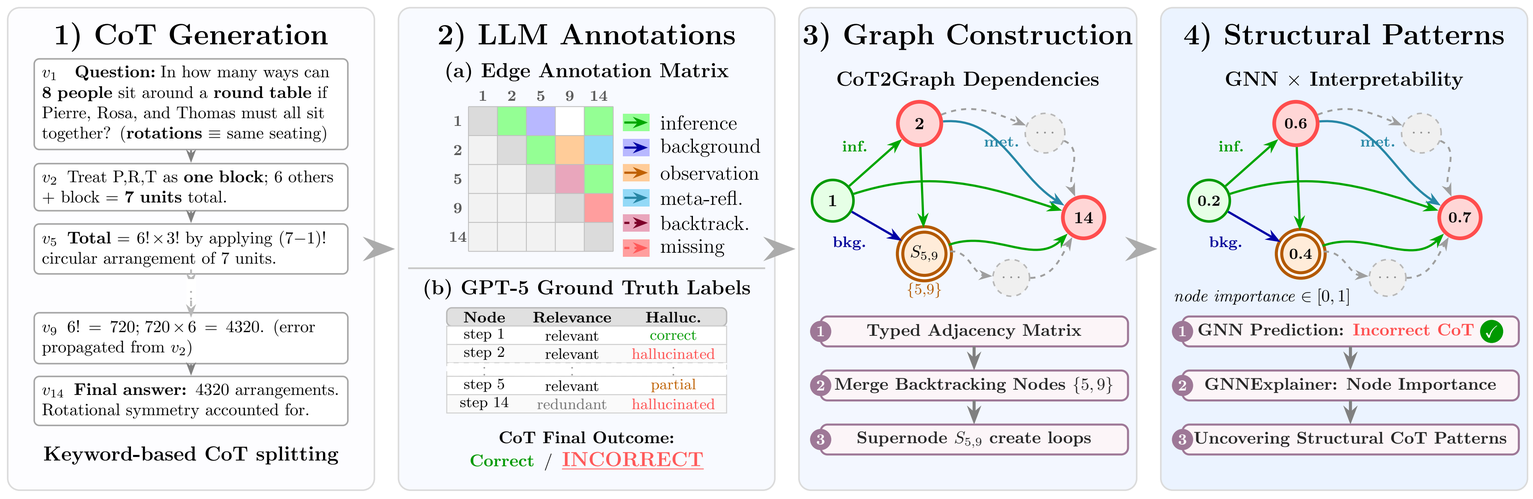
Long CoTs **backtrack**, **branch** and **merge**, but sequences and trees miss multi-source dependencies and revisits.

Q. Which aspects of reasoning quality are encoded in the **structure** of a CoT?



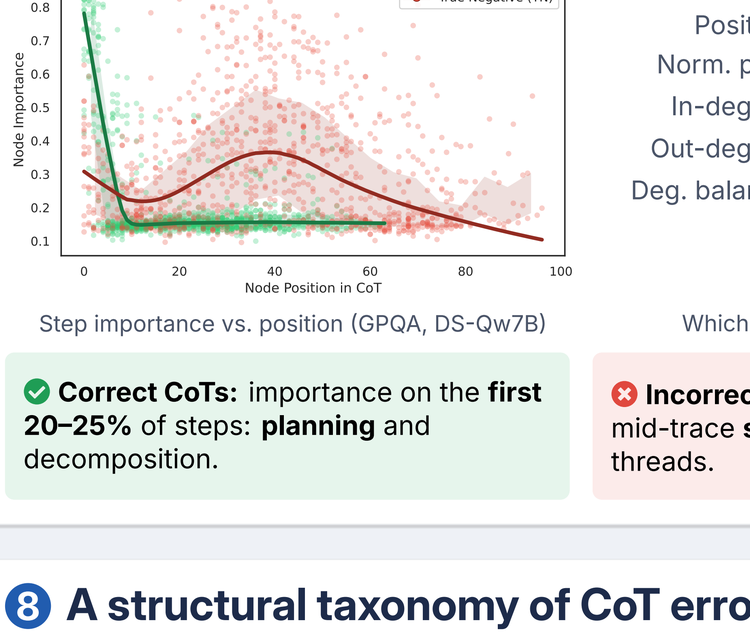
Nodes = reasoning steps · edges = typed dependencies

2 CoT2Graph: from reasoning text to a typed dependency graph



LLMs: DS-Qw7B, DS-Qw14B, MiMo-7B-RL on GPQA & MMLU-Pro · **Edges:** Llama-3.1-8B, 5 types + strength
Labels: GPT-5 Mini judge · **Model:** GATv2 on **structure only**, no step text

3 Predicting correctness

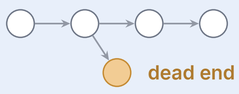


Graphs beat trees for every generator (up to +2.5%) and length by up to +7.8%.

Graph size and density alone score *below* the length baseline: the gain is structural.

4 What CoT structure can see

Redundancy lives *between* steps



66.5% accuracy on redundant steps **visible in the topology**

Hallucination lives *inside* a step

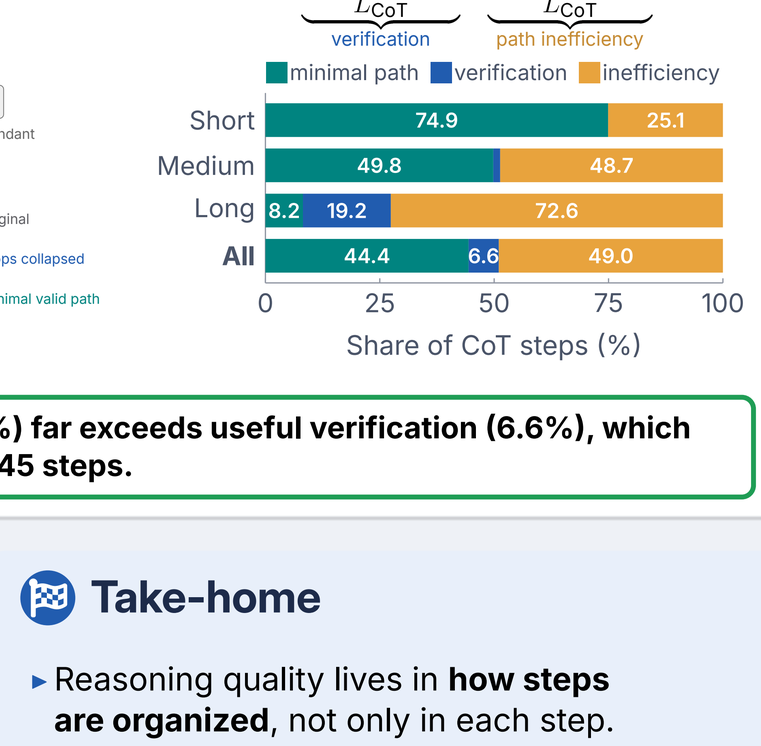


7.9% accuracy on hallucinated steps **invisible to structure**

Topology reveals redundant steps; hallucinations need semantic signals.

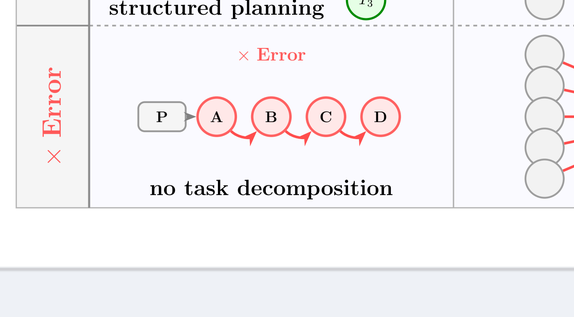
+7% more **skipped prerequisites** in hallucinated than in correct steps

5 Best-of-N selection

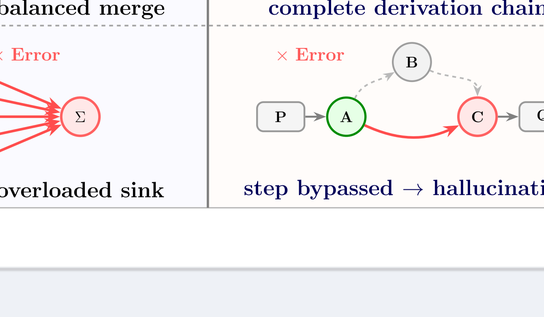


Best standalone verifier (+2.9 over the best baseline); with entropy it reaches 56.1%.

6 What separates correct from failed reasoning?



Step importance vs. position (GPQA, DS-Qw7B)



Which node features drive importance

Correct CoTs: importance on the **first 20–25%** of steps: **planning** and decomposition.

Incorrect CoTs: importance shifts to mid-trace **synthesis sinks** merging many threads.

7 Is long-CoT redundancy useful verification?

$$\mathcal{L} = \underbrace{\frac{L_{\text{CoT}} - L_{\text{loops}}}{L_{\text{CoT}}}}_{\text{verification}} + \underbrace{\frac{L_{\text{loops}} - L_{\text{TSP}^*}}{L_{\text{CoT}}}}_{\text{path inefficiency}}$$

Path inefficiency (49%) far exceeds useful verification (6.6%), which appears only after ∼35–45 steps.

8 A structural taxonomy of CoT errors

Take-home

- Reasoning quality lives in **how steps are organized**, not only in each step.
- Structure and uncertainty are **complementary** signals.
- Move toward **structure-aware evaluation** of reasoning.

Code & data
github.com/BanfiTommasoFelice/LLM-StructReasoning-EMNLP26